

PRE-INTERVIEW TAKE-HOME ASSIGNMENT

Build a Country Package approval service.

This coding exercise is for a Solutions Architect supporting World Bank Group country teams and the CPF/CPIA product portfolio. We are interested in pragmatic implementation choices, code quality, secure integration design, and how you explain the trade-offs you made.

FORMAT

Source repository + README

DISCUSSION

15-minute code walkthrough

The situation

Country teams produce **Country Packages** - structured artifacts that are reviewed, approved, and distributed to stakeholders at different organizational levels. Each package follows a pre-defined approval roadmap.

Build a service that supports the approval process for a country artifact. The service will later be integrated into a wider platform, so it should be clear, secure, testable, and ready to operate.

Technical brief

- Users have one of two roles: Country Editor or Country Reviewer. The role is scoped to one or more country codes.
- Build a REST API. Swagger/OpenAPI or equivalent API documentation is expected. A simple UI for testing the solution is considered a bonus.

- Use .NET with ASP.NET Core and C#. A relational database is preferred; PostgreSQL, SQL Server, SQLite, or an in-memory substitute is acceptable for the exercise.
- Use fictional data only. You may make reasonable assumptions and should document them.

Your task

For this exercise, you will implement a simplified Country Package approval system. The workflow consists of **four sequential steps**, each with a consistent interface: the Country Editor uploads a document, selects an approver (whose role is pre-defined for that step), and submits for approval. The Country Reviewer then logs in and acts on the submission. The system must enforce role-based access - only Country Editors may submit, and only Country Reviewers with the required clearance may approve. Every action must be recorded in an audit trail visible within the step's interface.

The approval steps are the following:

- **Obtain decision for country package from country level management** - this is a decision step; the Country Editor selects the correct user ID of the approver and submits. The package is then pending approval. The Country Reviewer may approve the step to complete it, or return the package for revision, allowing the Country Editor to make changes and resubmit.
- **Distribute the approved package to the country level management** - this is an information step; the Country Editor selects the correct user ID for the recipient of the distribution. Upon submission the package is marked as distributed and the step is completed.
- **Obtain decision for country package from regional management** - this step is similar to the obtain decision on country level but the stakeholder is on a regional level.
- **Distribute package to regional level management** - similar to distribution on country level but on regional level.

Each obtain decision step creates a snapshot of the document. The document may be revised in subsequent obtain decision steps, but documents attached to previously completed steps must remain unchanged.

Include the following items in your solution:

- **Model the approval process roadmap.** Design a data model for the Country Package that represents the pre-defined approval process. It must support attaching documents to the

steps that require them, specifying the correct approver per step, and logging an audit trail for all relevant actions.

- **Expose core operations.** A user with the Country Editor role must be able to create the process roadmap, upload a document per applicable step, and submit a step for approval. A user with the Country Reviewer role must be able to access and act on the steps assigned to them. Validate inputs and return useful, consistent error responses.
- **Enforce authorization.** A Country Editor may create a process roadmap, upload documents to it, and submit steps for review. A Country Reviewer may perform the review, approve a step, or return the package for revision - enabling the Country Editor to make changes and resubmit. Implement authentication in a way that makes these rules testable; a development-only header or local token is acceptable.
- **Test the important behavior.** Add automated tests for the key user flows and edge cases so critical behavior stays reliable as the code changes.

Bonus: Currently, Country Editors manually prepare and upload documents at each obtain decision step. Assuming a repository of unstructured data in the form of .pdf and .docx documents exists, design a solution that assists the Country Editor in generating these documents from the data repository — reducing manual effort and improving consistency across Country Packages.

Submission requirements

Required A solution architecture design document covering the data model, component diagram, service integrations, and RBAC approach across country codes and organizational levels.

Required A repository with runnable source code, automated tests, and a concise README.

Required Clear instructions to run the service and tests, plus sample requests or an API collection.

What we will look for

- Clear, maintainable code and appropriate separation of API, application, persistence, and integration concerns.
- Correct authorization, input validation, and thoughtful handling of sensitive operational data.
- Robust integration behavior, especially idempotency, partial failures, and preservation of user decisions.
- Tests that give confidence in critical behavior, not simply code coverage.
- Practical operational concerns: observability, diagnosability, configuration, and deployment readiness.
- Sound engineering judgment and clear explanations of the trade-offs in your implementation.

Submission process

Please share a repository link or compressed source archive at least 24 hours before the interview. During the interview, walk us through the code, key tests, and the most important trade-offs; we will use the remaining time to explore changes and operational scenarios.

Use public information or fictional examples only. Do not include confidential or personal data. There is no expected “right answer”; we value a small, working, well-reasoned solution over scope or visual polish. You may use existing tools available in the market to accelerate the implementation process.

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